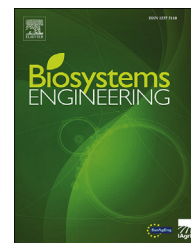


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Research Paper

Detection of the respiratory rate of standing cows by combining the Deeplab V3+ semantic segmentation model with the phase-based video magnification algorithm



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The respiratory rate (RR) of dairy cows is closely related to heat stress and reproductive diseases. In this study, a method was proposed to detect the RR of standing resting dairy cows by using deep learning and video analysis. First, a Deeplab V3+ semantic segmentation model was developed using the framework of ResNet-101, and the model was trained and tested using 3,000 manually labelled images to realise the segmentation of the cows. Next, a phase-based video magnification (PBVM) algorithm was applied to amplify the weak breathing movements of the cow. Finally, the Lucas-Kanade (LK) optical flow algorithm was used to detect breathing direction, and an RR detection model was used to detect the RR. The effectiveness of the method was verified by testing a total of 67,812 frames of data collected from 70 video segments of different cows. A comparison with the original video LK optical flow method (Method 1), target segmentation video LK optical flow method (Method 2), and original video motion amplification LK optical flow method (Method 3) showed that the accuracy of the proposed method was between 80.00% and 100.00%, and the mean accuracy was 93.04%. These values were 45.82%, 17.72%, and 22.94% higher than those of the other three methods, respectively. In addition, the method was tested in 105 videos of cows under heat stressed, and the RR detection accuracy was 98.69%. The results showed that the method could provide a reference for the automatic detection of RR and remote diagnosis of respiratory-related diseases.

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Nomenclature

ABS	Absolute value
ACC	Respiratory rate detection accuracy
Accuracy	Accuracy of segmentation
EVM	Eulerian video magnification algorithm
IOU	Intersection Over Union
$I(x, y, t)$	Gray value of pixel (x, y) at time t
I_x	Optical flow in the horizontal direction
I_y	Optical flow in the vertical direction
I_t	Changes of optical flow with time
LK	Lucas-Kanade optical flow algorithm
LSTM	Long Short-Term Memory
ToF	Time of flight
x, y	Location of pixel
u	Optical flow in the horizontal direction
m	Number of cow respiratory cycles
n	Number of pixel neighbourhoods
PBVM	Phase-based video magnification algorithm
P_{ii}	Number of pixels segmented correctly
P_{ij}	Number of pixels incorrectly segmented
P_n	Breathing direction of frame n
Prediction	Result of Deeplab V3+ segmentation
R_p	Respiratory rate detected by the proposed method
R_r	Actual respiratory rate
RR	Respiratory rate
Target	Real category of manually labelled pixels
v	Optical flow in the vertical direction
α	Angle between the average optical flow and the horizontal direction
θ	Angle between pixel's optical flow and horizontal direction

1. Introduction

The Respiratory rate (RR) of a dairy cow is closely related to the state of its physiological health. Automatic detection of RR of dairy cows that have been standing for a long time after eating is helpful for the evaluation of their physiological health and treatment of diseases (De Rensis, Garcia-Ispierto, & López-Gatius, 2015; Shwartz, Rhoads, VanBaale, Rhoads, & Baumgard, 2009; Silanikove, 2000). Because standing resting cows have weak respiratory movements, along with intermittent shaking, other body movements, and occlusal disturbance, detecting breathing movements is difficult, which hinders RR detection based on video analysis. Previously, the detection of RR of dairy cows mainly relied on the visual observation of cow abdominal fluctuations and RR estimation according to experience; however, such measurement is laborious and unsuitable for large-scale farming. Conversely, automated detection of RR is potentially of considerable significance for remote diagnosis and reducing the elimination rate of cows in large-scale farms (He, Liu, & Zhao, 2016).

Many studies have successfully developed automated RR detection methods. These methods can be mainly divided into

two types: contact-sensor and non-contact-sensor-based methods (AL-Khalidi, Saatchi, Burke, Elphick, & Tan, 2011; Jeong, Bychkov, & Searson, 2018; Yan, Liu, Jia, Tian, & Pi, 2017). The former methods rely on high-precision contact sensors to detect RR; however, contact sensors can cause stress reactions in dairy cows, and are vulnerable to scratching, water infiltration, weather changes and other factors that can damage the components and affect the service life of the sensor. Video analysis technology has the following advantages: low cost, rapid response, and no stress reactions in cows; it is an important development trend in the monitoring of the breathing rate of dairy cows (Bruyère et al., 2012; Cangar et al., 2008; Porto, Arcidiacono, Anguzza, & Cascone, 2013). Ji, Zhu, Liu, Li, and Ma (2011) applied machine vision to track the fluctuations of the ridge-abdomen contour of standing pigs and obtained the respiratory frequency by counting the fluctuations. In ten test groups, the respiratory detection accuracy of this method was 85.00%. Tang, Zhu, Li, and Yang (2015) used machine vision technology to detect the breathing rate of pigs by assessing the changes in their abdominal area during breathing. The results from ten test videos with low interference showed that the RR detection accuracy of this method could reach 92.0%. Xie, Ji, Hu, Yang, and Shen (2016) detected the RR of pigs by extracting the maximum curvature radius of their back contour in each video frame and constructed a respiratory waveform chart according to the extracted curvature radius. The average relative error between their method and manual detection results was 2.28%. Tao et al. (2017) obtained the depth images of pigs by using Kinect and extracted the abdominal region of pigs by using dynamic region detection and density clustering algorithm. The RR of pigs was determined by detecting the changes in the abdominal region depth with time. Compared with manual counting results, this method yielded RR detection accuracy of 85.3%. This method was found to be a feasible alternative for RR detection. Upadhyaya, Chatterjee, Prathosh, and Praveena (2016) proposed a human RR estimation method by combining depth learning technology and optical flow algorithm. First, the optical flow of thoracic motion during inhalation and exhalation was extracted in each video frame and input into the LSTM model and then the respiratory signal was estimated by using the trained LSTM model. Finally, the estimated respiratory signal was used to calculate the human RR. The correlation between the RR obtained by this method and the actual RR could reach 0.9, indicating that combining deep learning technology with traditional methods was feasible for detecting RR. Stewart, Wilson, Schaefer, Huddart, and Sutherland (2017) proposed a method for detecting RR in dairy cows based on infrared thermal imaging technology. A thermal infrared camera was used to monitor the temperature changes in the air near a cow's nostrils to detect its respiration. The mean error of the method was approximately $0.26\text{--}1.40\text{ breath min}^{-1}$, and the results of this method were highly consistent with these of manual counting. Zhao, He, and Wang (2014) proposed a method to detect the RR of a side-lying dairy cow according to the characteristics of the fluctuation of their abdomen during breathing. First, the Horn-Schunck optical flow method was used to detect the changes in the abdominal optical flow when the dairy cows breathe, and then the Otsu algorithm was used to screen out the

respiratory movement points. The RR of the dairy cows was detected by monitoring the change frequency of the direction of the optical flow at the respiratory movement points. After 72 tests, the RR detection accuracy of this method was 95.68%.

Respiration detection could be effectively realised by using a depth camera, microwave radar, time-of-flight (ToF) camera, and thermal infrared camera, but the image data widely used in dairy farms are mainly from ordinary RGB cameras (Lee, Pathirana, Evans, & Steinfert, 2015; Penne, Schaller, Hornegger, & Kuwert, 2008). Moreover, since cows are large and the changes in the ridge-abdomen contour are not obvious, these features cannot be reliably used for breath detection of standing cows. Considering the complex environment and problems with noise on dairy farms, using the acoustic Doppler effect or an ultrasonic proximity sensor for breath detection is difficult (Wang et al., 2018). The noise mainly includes cow motion interference and other acoustic noise. Although the Horn-Schunck optical flow algorithm could be used to detect the breathing rate of laterally lying cows with high accuracy, it involves a dense optical flow with a large amount of computation. In addition, standing cows have a weak breathing movement range, rendering the detection of the RR of standing cows more difficult (Bruhn, Weickert, & Schnörr, 2005; Gong & Bansmer, 2015). The reason why the resting RR was detected was that the resting RR of standing dairy cows could better reflect their physiological health status, which was more instructive for disease diagnosis based on this technology (Song, Wu, Yin, Jiang, & He, 2019; Zhao et al., 2014).

This study proposes the detection of the RR of standing resting cows by combining the deep learning and video analysis methods. The removal of irrelevant background details and extraction of the cow target was realised by segmenting video sequence frames, and then amplifying the cow's breathing movements. Finally, the LK sparse optical flow algorithm was used to detect the average optical flow direction of each frame in the video, and the RR of the standing resting dairy cows was detected and counted using an RR detection model. This method can provide a reference for the accurate and automatic detection of RR in standing resting cows and allow the automated monitoring and remote diagnosis of diseases related to breathing in cows, especially heat stress.

2. Materials and methods

2.1. Materials

The experimental video material was collected from a commercial dairy farm in Shaanxi Yangling, China, between April and May 2019. The videos were recorded from 2:00 p.m. to 5:00 p.m., when most of the cows were standing and resting. The subjects were standing Holstein cows in the middle of lactation and at rest in a sports field. As shown in Fig. 1, during this resting period, a Sony HDR-CX290 camera (Sony Co., Tokyo, Japan) was fixed on a tripod at the side of the sports field at a distance of about 3 m from the cow. The camera's viewing direction was perpendicular to the cow's standing direction, and the frame rate/code rate was 50 fps/2000 kbps, and the

resolution was 720 pixels (horizontal) $\times$ 576 pixels (vertical). The camera was adjusted so that the cow's abdominal area accounted for about 80.00–90.00% of the screen, because the regular movement of a cow's abdomen is closely related to its state of respiration.

In this study, a total of 22 cows were monitored in different environments and with different interference factors. Five videos were obtained for each cow, and a total of 110 videos each lasting about 20 s were recorded. All the videos were captured from standing resting cows.

From the 110 videos, 40 segment videos were randomly selected and decomposed into video sequence frames; from these, 3000 images of cows were randomly selected. The cow target and irrelevant background in the images were marked manually, and these images were used as training and test data for the Deeplab V3+ target segmentation network (70% were used as training data and 30% as test data). In order to avoid other cows blocking the view of the cow target, to achieve a better segmentation of the target cow, and to improve the robustness of the proposed method, the nearest cow in a group was marked as the cow target, and the other interfering cows and the background were considered as background. The remaining 70 videos were used for the verification of the proposed RR detection method. To verify the effectiveness of the proposed method in detecting RR under heat stress as an example of an important physiological stress, an additional 105 test videos (RR between 90 and 222 breath min^{-1}) under different weather (including heat stress conditions), wind speed, movement and occlusion interferences were evaluated. These videos were obtained in a previous study and full details are given by Song et al. (2019).

Table 1 gives detailed information on the 70 test videos, including the video number, shooting date, and duration and total number of frames; different weather parameters, wind velocity and interference factors such as flying insects, near-view targets, other dairy cows and body shaking. These factors increase the difficulty of detecting RR in standing dairy cows, but they can also be used to validate the robustness of the method. The flying insects mainly include mosquitoes and flies. The tail minor movement mainly refers to the tail movement driven by wind or body shaking, whereas tail strenuous movement is controlled subjectively by dairy cows, has a large amplitude and can last for a long time. The legs minor movement refers to leg movement when the hoof does not leave the ground, and strenuous movement is leg movement when the hoof frequently leaves the ground (Song, Li, Jiang, Wu, & He, 2018; Zhao et al., 2014).

The processor used in the video processing platform was an Intel Core E5-1620, with a 3.50 GHz processor, 32 GB RAM, 500 GB hard disk and 5 GB NVIDIA Quadro P2000 GPU, and the algorithm was developed under MATLAB 2019a environment.

2.2. RR detection of standing resting cows by using deep learning and video analysis

2.2.1. Overall technical approach

The approach used to detect the RR of standing resting cows is shown in Fig. 2. The proposed method consists of three stages: segmentation of the cow target, amplification of the breathing

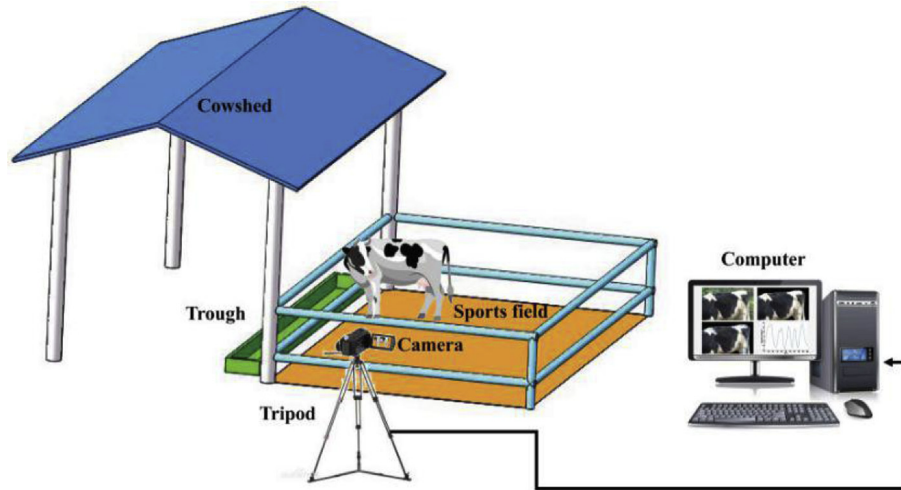


Fig. 1 – Schematic diagram of video capture process.

Table 1 – Information of test videos.

Video No.	Shooting date	Weather	Wind Velocity m s^{-1}	Duration/s	Number of frames	Interference factors
1–3	11 April & 17, 23 May	sunny	1.6–3.3	20	1000	Background strenuous movement
4–5	9 May	sunny	0.3–1.5	20	1000	Head minor movement
6–8	6 April	sunny	0.3–1.5	20	1000	Head minor movement
9	6 April	sunny	0.3–1.5	19	945	Head minor movement
10–12	17, 22 April & 15 May	cloudy	1.6–3.3	20	1000	Head minor movement
13–15	6 April	sunny	1.6–3.3	20	1000	Head strenuous movement
16–17	11 April & 9 May	sunny	1.6–3.3	20	1000	Head strenuous movement
18	8 April	cloudy	3.4–5.4	19	963	Head strenuous movement
19–23	23 April	sunny	0.3–1.5	20	1000	Body minor movement
24–28	5, 6, 23 April & 9, 17 May	sunny	1.6–3.3	20	1000	Body minor movement
29–30	25 April	sunny	1.6–3.3	20	1000	Body minor movement
31–33	9, 24 April & 21 May	cloudy	1.6–3.3	20	1000	Body minor movement
34–35	18 April & 17 May	sunny	1.6–3.3	20	1000	Body strenuous movement
36–38	18 April	sunny	1.6–3.3	20	1000	Body strenuous movement
39–41	30 April	cloudy	3.4–5.4	20	1000	Body strenuous movement
42–44	12 May	cloudy	3.4–5.4	20	1000	Legs minor movement
45–46	9 May	sunny	0.3–1.5	20	1000	Flying insects
47–50	5, 6, 11 April & 9 May	sunny	1.6–3.3	20	1000	Flying insects
51	17 May	sunny	1.6–3.3	20	1000	Flying insects
52	4 May	cloudy	0.3–1.5	19	949	Tail minor movement
53–56	4, 9, 14 April & 8 May	cloudy	1.6–3.3	20	1000	Tail strenuous movement
57–59	15 May	cloudy	1.6–3.3	20	1000	Tail strenuous movement
60–62	6 April	sunny	0.3–1.5	20	1000	Near-view targets
63–67	16 April	cloudy	1.6–3.3	20	1000	Near-view targets
68–69	23 May	sunny	1.6–3.3	20	1000	Other dairy cows
70	26 May	cloudy	1.6–3.3	20	955	Other dairy cows

movement, and RR detection. In the first stage, the Deeplab V3+ model, a semantic segmentation model constructed using fine-tuning ResNet-101, was designed to achieve the segmentation of the cow target and removal of the irrelevant background, in order to reduce the effects of background noise on the amplification of the breathing movement of the cow. In the second stage, the PBVM algorithm was used to amplify the respiratory movement of the standing resting cow to facilitate RR detection. In the third stage, the LK sparse optical flow algorithm was used to detect the average optical flow direction of each video frame, and the RR detection model is used to detect and count the RR of the standing resting cow.

2.2.2. Dairy cow target segmentation based on Deeplab V3+
In order to avoid interference from irrelevant background, target occlusion, and near-view targets as well as to improve the robustness of the RR detection algorithm, we first segmented the cow target and removed the irrelevant background. Since the environmental background of standing cows is more complex than that of lateral lying cows, and there are many interference factors, the Deeplab V3+ algorithm was selected for this study because of its high precision and robustness in deep semantic segmentation (Lateef & Ruichek, 2019; Lim, Cheng, Hsu, & Lee, 2016; Morales et al., 2018).

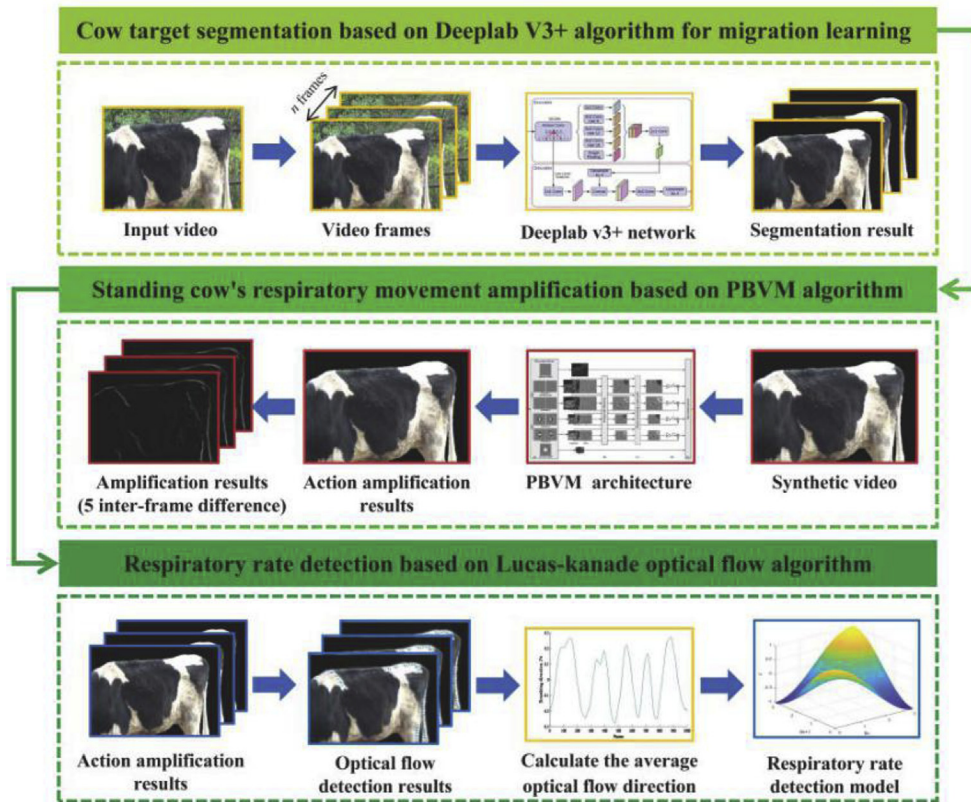


Fig. 2 – Technical approach of the proposed method.

Deeplab V3+ is based on Deeplab V3; it introduces an encoder-decoder module that is commonly used in semantic segmentation. It can arbitrarily control the resolution of feature extraction by an encoder and can improve the segmentation effect of the target boundary (Chen, Zhu, Papandreou, Schroff, & Adam, 2018). As shown in Fig. 3, the Deeplab V3+ model in this study was designed using ResNet-

101 and trained on the ImageNet database as the skeleton; the training and testing of Deeplab V3+ were completed by manually labelling the images of standing cows to achieve segmentation of the target and irrelevant background. At the encoder stage, the complete connection layer and all subsequent layers in ResNet-101 were first removed to complete feature extraction, and the feature map was then input into

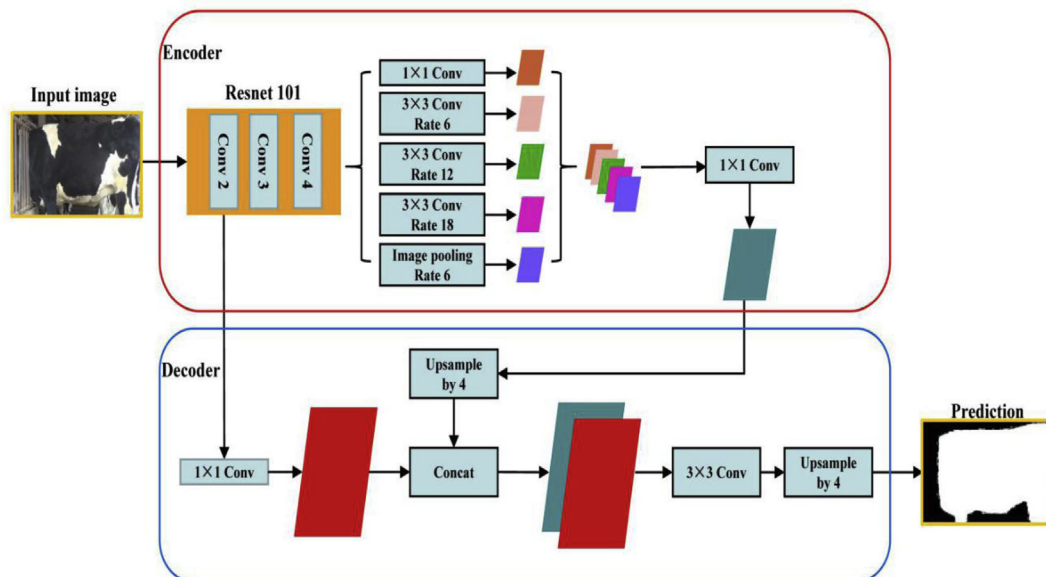


Fig. 3 – Deeplab V3+ network structure based on ResNet-101 skeleton.

the atrous spatial pyramid pooling (ASPP) structure. The output structure of ASPP was then converted to a channel number by using the 1×1 convolution layer. The feature map was input into the global pooling layer for feature extraction, and global average pooling was used to obtain image-level features. At the decoder stage, the lower feature maps in the ResNet-101 feature extraction network were first selected (the Conv2 feature map was selected in this study). After transforming the channel was transformed using a 1×1 convolution operation, the feature map and the encoder output were stacked at the same size, and the target segmentation was then completed using a 3×3 convolution operation and up-sampling for four times. The main purpose of the 1×1 convolution was to adjust the number of channels, and prevent the prediction results from tilting to low-level features. The purpose of the 3×3 convolution operation was to learn and increase non-linearity, so that the low- and high-level features could be more effectively combined, whereas the upper sampling aimed to increase the size of the feature map. The ASPP structure consisted of a 1×1 convolution, three 3×3 void convolutions, and three void convolutions, with void ratios of 6, 12, and 18, respectively. All convolutions had 256 convolution cores, and after processing, the output characteristic graphs contained 256 channels.

The training parameters for the constructed Deeplab V3+ model on the manually labelled standing cow target data set were those used by Chen et al. (2018). The specific parameters are shown in Table 2.

The 3,000 manually labelled images were divided into a training set (2,100 frames) and a test set (900 frames), i.e. a 7:3 ratio, to complete the training and testing of the model. The accuracy curve and loss curve for the training process are shown in Figs. 4 and 5, respectively.

The accurate segmentation of cow target from the complex background is of considerable significance for the subsequent detection of cow RR. *Intersection over Union* (IOU) and *Accuracy* are important evaluation indexes in semantic segmentation (Yang et al., 2017). As shown in Eqs. (1) and (2), in order to measure the segmentation effect of Deeplab V3+ model constructed in this study, we used the trained model to complete cow target segmentation of the test set and selected IOU and Accuracy to measure the result of the target segmentation.

$$IOU = \frac{Target \cap Prediction}{Target \cup Prediction} \quad (1)$$

$$Accuracy = \frac{\sum_{i=0}^k P_{ii}}{\sum_{i=0}^k \sum_{j=0}^k P_{ij}} \quad (2)$$

where *Target* is the real category result of manually labelled pixels, and *Prediction* is the result of Deeplab V3+ segmentation. P_{ii} is the number of pixels belonging to class i where the result of segmentation is class i , whereas P_{ij} refers to the number of pixels belonging to class i where the result of segmentation is class j .

The test results are shown in Tables 3 and 4. The Accuracy of segmentation results and the IOU are shown in Table 3. The confusion matrix of the test results is shown in Table 4. The results suggested that the Accuracy of the segmentation of the cow target reached 99.40%, and the IOU reached 0.987, indicated that the proposed Deeplab V3+ model based on ResNet-101 could achieve high-precision segmentation of standing cow targets.

The segmentation results are shown in Fig. 6. Among them, the cow target on one side of the cow sports field in Figs. 6 b-1 is standing resting, which looks like urinating due to a wooden post at the back. The target segmentation was relatively integral, with accurate contour positioning and clear boundaries, indicating that it is an effective and feasible method of achieving target segmentation of standing cows, thus laying the foundation for the next step of amplifying breathing movement. The main reason for some background pixels on the back of the cow in Figs. 6 a-2 was that it was difficult to completely and accurately label the cow target and background in the model training data set, which caused errors in the model training, which would have a certain impact on the final segmentation results. The two cow targets with similar colours in Figs. 6 c-2 could be well segmented because we marked the interfering cow as the background instead of the cow target when generating the data set, which allowed the trained model to adequately handle the occlusion problem.

A video is available as supplementary data relating to the segmentation results in this manuscript when there is occlusion interference for other cows:

Supplementary video related to this article can be found at <https://doi.org/10.1016/j.biosystemseng.2020.01.012>.

Table 2 – Parameters setting for the training of Deeplab V3+ model.

Parameter name	Value
GPU	5 GB NVIDIA Quadro P2000
Initial learning rate	1.00×10^{-3}
Learning Rate Updating Strategy	Update learning rate every 2 epochs
Weight decay	5×10^{-5}
Batch size	1
Momentum	0.90
Iteration	21,000
Gradient descent strategy	Momentum

2.2.3. Standing resting cow's respiratory movement amplification based on the PBVM algorithm

Since standing resting cows show weak respiratory movement and this is affected by numerous interference factors, detecting their breathing rate is difficult. Therefore, amplifying the respiratory movement is important. Based on the completed target segmentation, we utilised the PBVM algorithm to amplify this weak respiratory movement.

PBVM is a phase-based video magnification algorithm based on traditional Eulerian video magnification (EVM) technology; it effectively removes the 'ghost shadow' generated in EVM. The algorithm flow chart is shown in Fig. 7 (Chen et al., 2015; Wadhwa, Rubinstein, Durand, & Freeman, 2013; Wu et al., 2012).

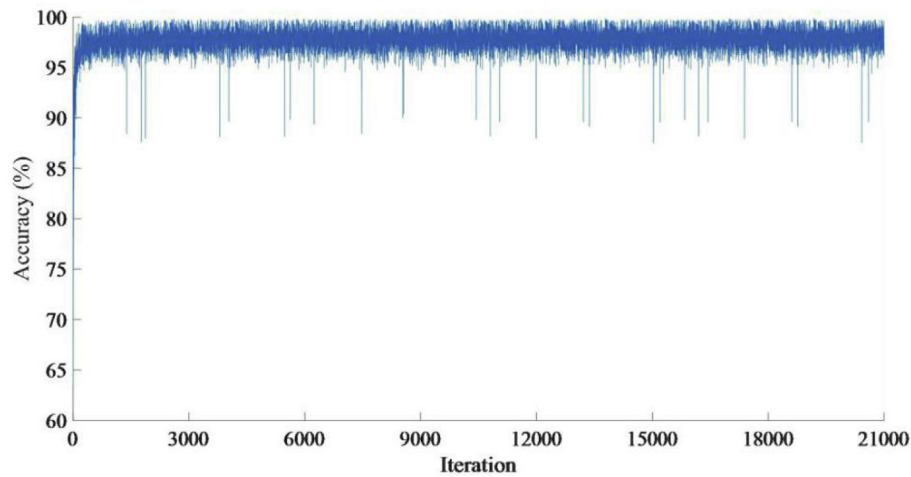


Fig. 4 – Training accuracy curve of the Deeplab V3+ network.

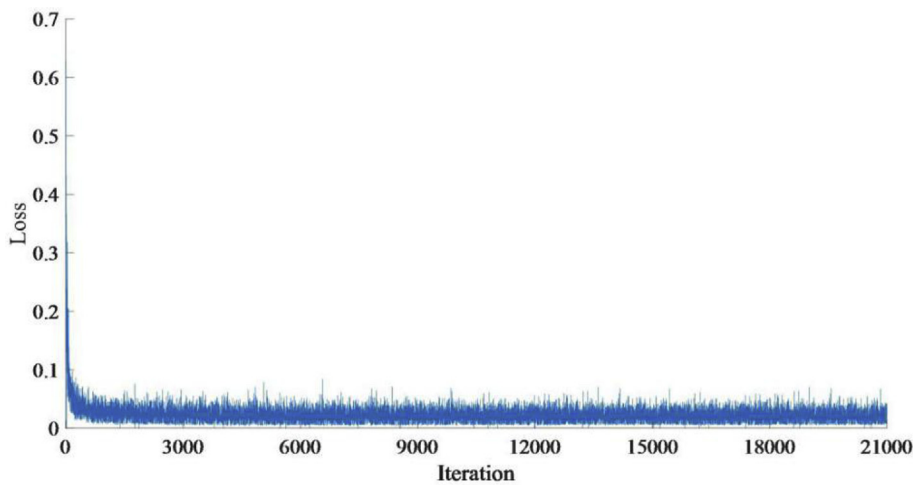


Fig. 5 – Training loss curve of the Deeplab V3+ network.

The algorithm consists of the following main steps:

- (1) Space pyramid transformation: The input video is first spatially decomposed using a complex manipulable pyramid to extract the amplitude values and phases of the images at different scales and in different directions.

- (2) Time domain band-pass filter: Sub-bands with different scales and directions are subjected to a filtering processing within the time domain required to remove the direct current component by using a band pass filter.
- (3) Motion amplification: The filtered phase is changed to amplify the motion amplitude and reconstruct the image.
- (4) Video synthesis: The pyramid is rebuilt and the video is output.

Table 3 – Cow target segmentation results.

	Accuracy	IOU
Cow	0.994	0.987
Background	0.960	0.929

Table 4 – Confusion matrix of cow target segmentation results.

	Cow	Background
Cow	0.994	0.006
Background	0.040	0.960

The PBVM algorithm needs to specify the motion frequency (minimum and maximum frequencies) of the movement to be amplified, the magnification factor of the movement, the decomposition scale of the image, and the type of band-pass filter used. Since the abdomen of dairy cows moves regularly with the changes in respiratory state, according to the range of respiratory frequency, the motion frequency selected was approximately 0.30–3.30 Hz (Han, Liu, & Liu, 2018). The decomposition scale and magnification factor of 22 layers and 8, respectively, were used, as specified by Wadhwa et al. (2013). Standing resting cows

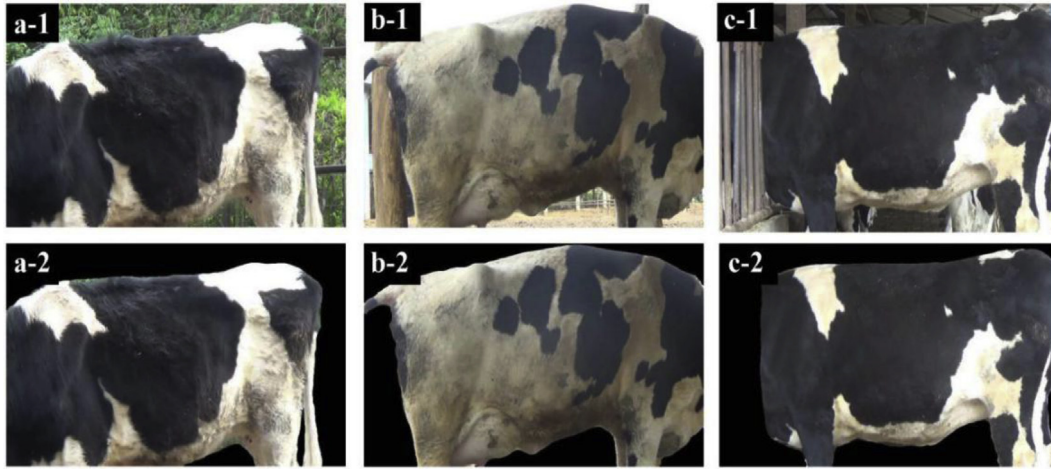


Fig. 6 – Target segmentation results of standing resting cows based on Deeplab V3+: (a-1) Video 1 is disturbed by complicated background; (b-1) Video 62 is disturbed by a near-view object; (c-1) Video 69 is disturbed by other cows; (a-2), (b-2) and (c-2) correspond to the segmentation results of (a-1), (b-1) and (c-1), respectively.

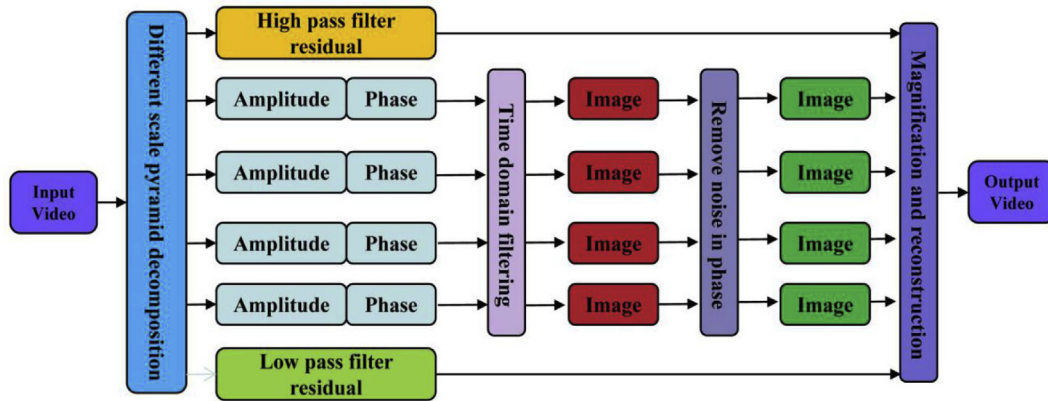


Fig. 7 – Process of movement amplification by PBVM algorithm.

have weak respiratory movement that is difficult to observe and detect, and since the IIR has a wide band-pass, which effectively extends and amplifies minute motions, it was selected as the band-pass filter in this study (Li & Chen, 2018; Park & Kim, 2009).

The results of respiratory movement amplification are shown in Fig. 8. Rows 3 and 4 of Fig. 8 reveal that, in the original video, observing the changes in the movement of cows over a 5-frame difference is still difficult, whereas these changes could be clearly seen after the PBVM algorithm has been applied, indicating that this algorithm can effectively amplify the weak movements of a standing resting cow. This algorithm forms the basis for detection of RR by detecting the body movements of a standing cow by using the LK sparse optical flow algorithm.

2.2.4. RR detection based on LK sparse optical flow algorithm

After target segmentation and breathing movement amplification, the body of the standing resting cow regularly reciprocated the changes in the breathing state. Horn-Schunck and LK optical flow algorithms are two commonly used optical

flow algorithms. Unlike the former, the LK algorithm is a two-frame differential optical flow estimation algorithm that processes fewer pixels and takes less time (Baker et al., 2011; Chao, Gu, & Napolitano, 2014; Jing, Qi, Yang, & Lu, 2018; Lee, Choi, & Park, 2007); therefore, we used this algorithm to calculate the average optical flow direction of each frame. Based on the changes in the direction, we used the detection model to detect and count the RR of the standing resting cow. The LK optical flow model is based on three assumptions: (i) the brightness must be constant, and the same point changes its brightness with time; (ii) movements are small, and changes over time do not cause dramatic changes in position; and (iii) the spatial position is consistent, that is adjacent points in the scene are projected to adjacent positions on the image, and the adjacent points are in the same speed (Lucas & Kanade, 1981; Perrone, 1990; Shukla & Patel, 2013). From condition (i), we know that:

$$I(x, y, t) = I(x + dx, y + dy, t + dt) \quad (3)$$

where $I(x, y, t)$ and $I(x + dx, y + dy, t + dt)$ are the grayscale values of pixel (x, y) and $(x + dx, y + dy)$ at time t and $t + dt$,

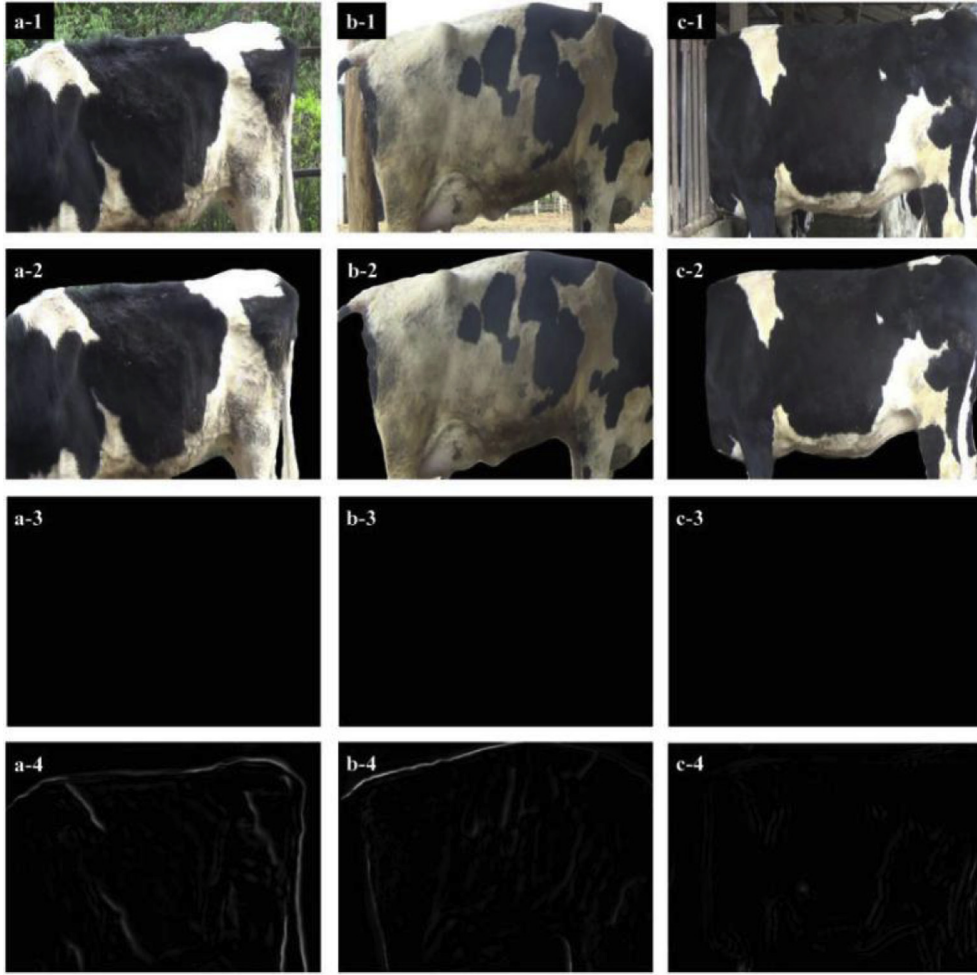


Fig. 8 – Results of respiratory motion amplification based on PBVM: (a-1) Video 1 is disturbed by complicated background; (b-1) Video 62 is disturbed by a near-view object; (c-1) Video 69 is disturbed by other cows; (a-2), (b-2) and (c-2) correspond to the segmentation results of (a-1), (b-1) and (c-1), respectively; (a-3), (b-3) and (c-3) are 5-frame differences of the segmentation results (a-2), (b-2) and (c-2), respectively; (a-4), (b-4) and (c-4) are the 5-frame difference results of segmentation results (a-2), (b-2) and (c-2) amplified by PBVM.

respectively. dt approaches zero indefinitely and the gray-scale of pixel (x, y) remains unchanged before and after the motion, that is, $\alpha I(x + dx, y + dy)/\alpha t = 0$. We can expand this to give Eq. (4):

$$(\alpha I / \alpha x) \cdot (dx / dt) + (\alpha I / \alpha y) \cdot (dy / dt) + \alpha I / \alpha t = 0 \quad (4)$$

We let $I_x = \alpha I / \alpha x$, $I_y = \alpha I / \alpha y$, $I_t = \alpha I / \alpha t$, and assume that u and v denote the optical flow in the horizontal and vertical directions of the pixel (x, y) , respectively. Since $u = dx/dt$, and $v = dy/dt$, then Eq. (4) can be abbreviated as Eq. (5):

$$I_x \cdot u + I_y \cdot v + I_t = 0 \quad (5)$$

From condition (iii), we know that the pixel (x, y) is the same as the pixel motion in the neighbourhood. To solve the optical flow information of the pixel (x, y) , we can establish a constraint equation in the neighbourhood of the target pixel. As shown in Eq. (6), the optical flows u and v are resolved and solved, and the horizontal optical flow u and the vertical optical flow v of the target pixel are obtained from Eq. (5).

$$\begin{bmatrix} I_{x1} & I_{y1} \\ I_{x2} & I_{y2} \\ \vdots & \vdots \\ I_{xn} & I_{yn} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} I_{t1} \\ I_{t2} \\ \vdots \\ I_{tn} \end{bmatrix} \quad (6)$$

The optical flows u and v of the target pixel (x, y) are calculated using the LK sparse optical flow algorithm, and the optical flow direction of the target pixel (x, y) (the angle between the resultant optical flow of the target pixel (x, y) and the horizontal direction θ) is then calculated using Eq. (7):

$$\theta = \arctan v/u \quad (7)$$

The RR can be calculated on the basis of the change in the average optical flow direction of the video sequence frame by calculating the direction α of the average optical flow of each frame image by using Eq. (8), where α represents the angle between the average optical flow and the horizontal direction, θ_n is the direction of optical flow in the n th pixel neighbourhood. The main reason for using the average optical flow

direction to detect the RR of dairy cows was that the average optical flow direction had better robustness (Song et al., 2019).

$$\alpha = (\theta_1 + \theta_2 + \dots + \theta_n) / n \quad (8)$$

We found that, when standing resting cows breathe, the body periodically reciprocated along the horizontal line as the breathing state changed, and this was reflected in the image as a horizontal reciprocating change in the direction of the optical flow of the cow target. Since complete respiratory cycles included two processes of exhalation and inhalation, the cow would have a horizontal reversal in the direction of its optical flow when it completes a breath. The calculation method of cow respiratory cycles number in the video is shown in Eq. (9), which can detect the change in the sign of the cosine value of the average optical flow direction of two consecutive video frames in order to detect the respiratory cycles.

$$\begin{cases} \cos \alpha_n \cdot \cos \alpha_{n+1} < 0 & \text{Increased } m \text{ by } 1 \\ \cos \alpha_n \cdot \cos \alpha_{n+1} \geq 0 & m \text{ remains the same} \end{cases} \quad (9)$$

where α_n , α_{n+1} , and m represent the average optical flow directions of the n th and the $n+1^{\text{th}}$ frames and the number of cow respiratory cycles in the test video, respectively. $P_n = \cos \alpha_n$, $P_{n+1} = \cos \alpha_{n+1}$, and P are called the respiratory directions. Eq. (9) can then be written as Eq. (10):

$$\begin{cases} P_n \cdot P_{n+1} < 0 & \text{Increased } m \text{ by } 1 \\ P_n \cdot P_{n+1} \geq 0 & m \text{ remains the same} \end{cases} \quad (10)$$

Considering that the respiratory cycle number m can be obtained from the test video, and the frame rate of test video is 50 fps, the RR detection model of test video is shown in Eq. (11) below:

$$R = \frac{50 \times m}{u} \quad (11)$$

where, R is the RR of the cow in the test video, 50 is the frame rate, and u is the total frame number of the test video.

The results of respiratory movement detection are shown in Fig. 9. It is difficult in Figs. 9a-2 and 9b-2 to detect the optical flow generated by the respiratory movement of standing resting cows due to the weak respiratory movement. Fig. 9c-2 shows that only the optical flow generated by the interference area can be detected when there is severe motion interference (leg movements) in the original video. Since the standing cow's respiratory behaviour is difficult to observe and the range of motion is small, it is also impossible to clearly detect the optical flow generated by the cow's breathing action. However, Figs. 9a-3, 9b-3 and 9c-3 show that after target segmentation and respiratory movement amplification, the LK sparse optical flow algorithm can be used to detect the optical flow generated by the body movements of the standing cows when breathing. Successful detection of the optical flow generated by a cow's breathing action lays the foundation for detecting the RR of a standing cow based on the respiratory direction.

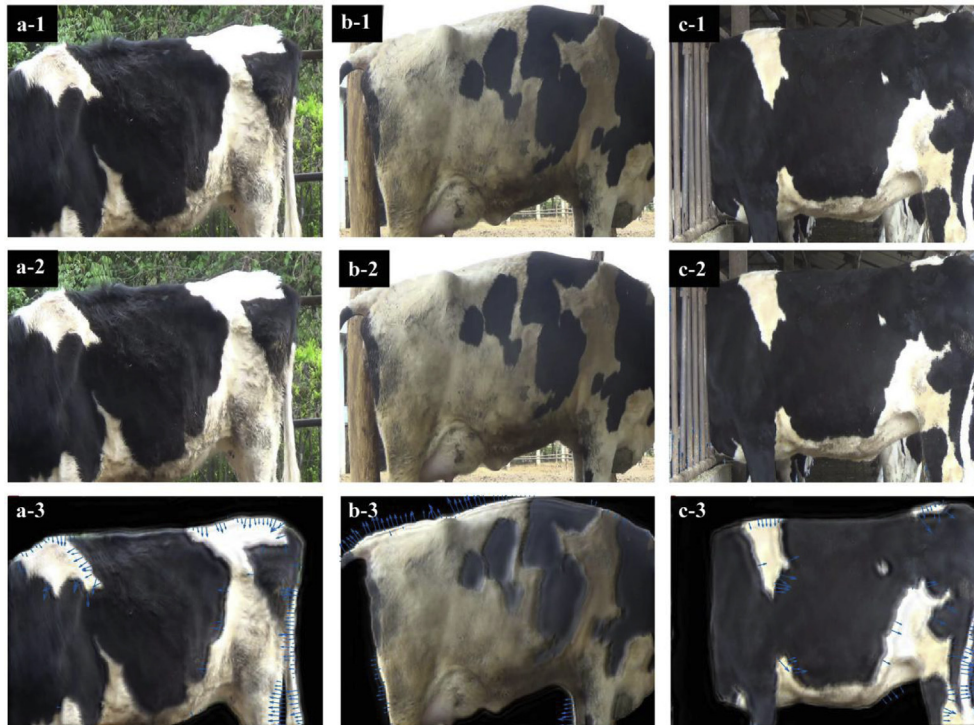


Fig. 9 – Cow's respiratory movement detection result based on LK algorithm: (a-1) Original Video 1 is disturbed by background; (b-1) Original Video 62 is disturbed by a near-view target; (c-1) Original Video 69 is disturbed by other cows; (a-2), (b-2) and (c-2) are (a-1), (b-1) and (c-1) optical flow detection results based on LK algorithm respectively; (a-3), (b-3) and (c-3) are respectively (a-1), (b-1) and (c-1) optical flow detection results by LK algorithm after target segmentation and respiratory movement amplification.

The detection results of the respiratory direction corresponding to Videos 1, 62, and 69 show the effects of disturbances (Fig. 10a-c). There were clear peaks and valleys in the changes in the breathing direction, and the changes in the respiratory state were clearly distinguishable. The proposed method was found to be effective for the detection of respiratory cycles in standing resting dairy cows, where, a positive and negative change in the value of the breathing direction indicates a breathing cycle.

2.3. Evaluation criterion

In order to verify the accuracy of the study method in detecting the RR of a standing resting cow, the accuracy ACC was utilised in this study.

The actual respiratory rate R_r is calculated according to Eq. (12) on the basis of the known number of test video frames u in combination with manual counting of the number of respiratory cycles in the test video. The main reason for selecting manual counting was that its error would only appear at the end of the video, and the error would not accumulate over time, and the maximum error is within one. The RR detected by the proposed method is R_p , and the breathing rate detection accuracy for the standing cows is shown in Eq. (12):

$$ACC = (1 - \text{ABS}(R_p - R_r) / R_r) \times 100.00\% \quad (12)$$

3. Results and analysis

The test results of the 70 test videos obtained using the proposed method are shown in Fig. 11, and the detection accuracy of the RR is shown in Fig. 12.

It can be seen from Fig. 11, Fig. 12 and Table 1 that the 70-segment test videos included 43 instances of sunny conditions and 27 of cloudy conditions. The average accuracy of RR detection was 92.86% on sunny days, and 93.31% on cloudy days, indicating that the proposed method successfully compensated for interference from illumination conditions. Since the test cows were in an unstructured environment, the effects of wind velocity on the test results needs to be considered. The influence of wind velocity on the detection results was verified using the variance of average detection accuracy under different wind velocities to measure the influence of wind speed changes on respiratory detection results. Of the 70 test videos, 17, 46, and 7 were recorded at different wind speeds (0.3–1.5, 1.6–3.3, and 3.4–5.4 m s⁻¹, respectively). The average accuracy of RR detection in these three environments was 93.83%, 92.62% and 92.64%, respectively, and the variance of the detection accuracy under the three wind speeds was 0.00005. This showed that the proposed algorithm was robust to wind under normal conditions. Of the 70 test videos, 69 had RR detection accuracy of more than 80.00%; 54, more than 90.00%; and 26, 100.00%. The average overall accuracy and RR detection error were 93.04% and 2.4 breath min⁻¹,

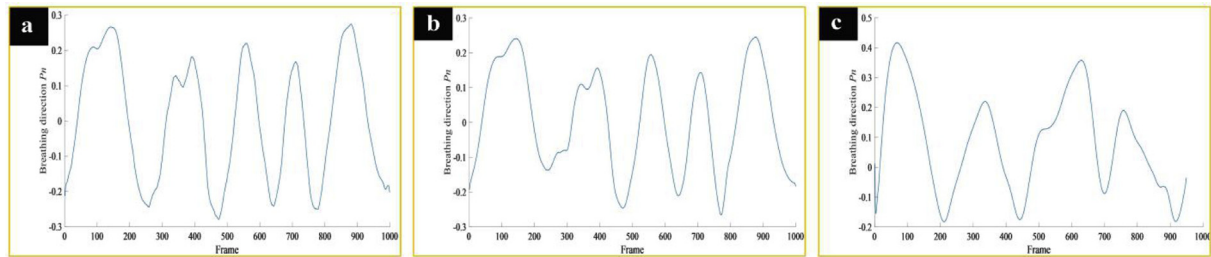


Fig. 10 – Respiratory direction test result: (a) disturbed by background (Video 1); (b) disturbed by a near-view target (Video 62); (c) disturbed by other cows (Video 69).

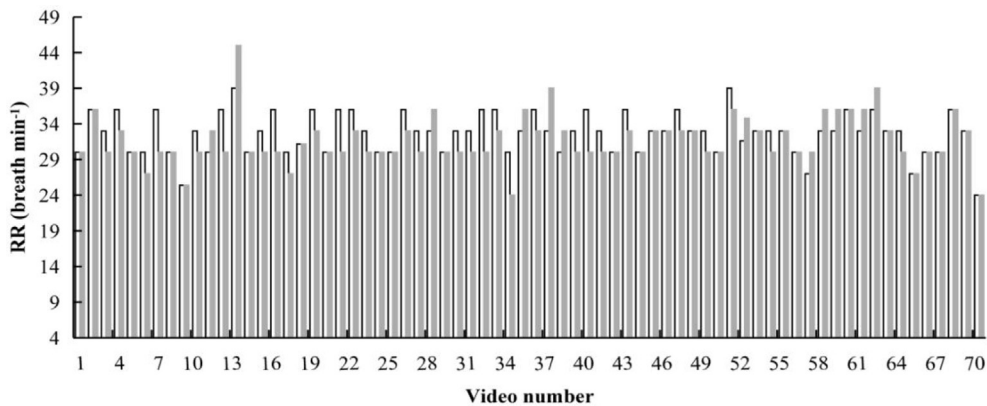


Fig. 11 – RR test results of the proposed method on 70 groups of test videos: white histogram and gray histogram respectively represent the actual RR of each video and the RR detected by the proposed method.

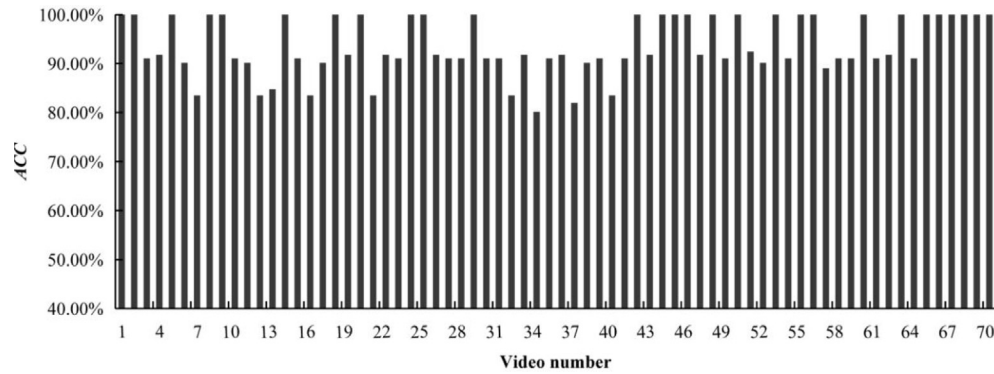


Fig. 12 – RR detection accuracy of the proposed method on 70 groups of test videos.

respectively. These results showed that the proposed method could effectively detect the RR of standing cows in a resting state.

3.1. Comparison with other detection methods

In this study, the proposed method was compared with Method 1, Method 2, and Method 3 to verify effectiveness. The results of breathing rate detection of standing cows by using

the three methods are shown in Fig. 13. The average ACC of RR detection by using Methods 1, 2 and 3 was 47.22%, 75.32%, and 70.10%, respectively. Their average detection error for RR was 17.4, 7.8, and 9.6 breath min^{-1} , respectively. The accuracy of Method 1 was the lowest and the error was the largest. The average error of Method 2 was 1.8 breath min^{-1} lower than that of Method 3, and its ACC was considerably higher than that of Method 3. The ACC of the method proposed in this study was significantly higher than any of the other three

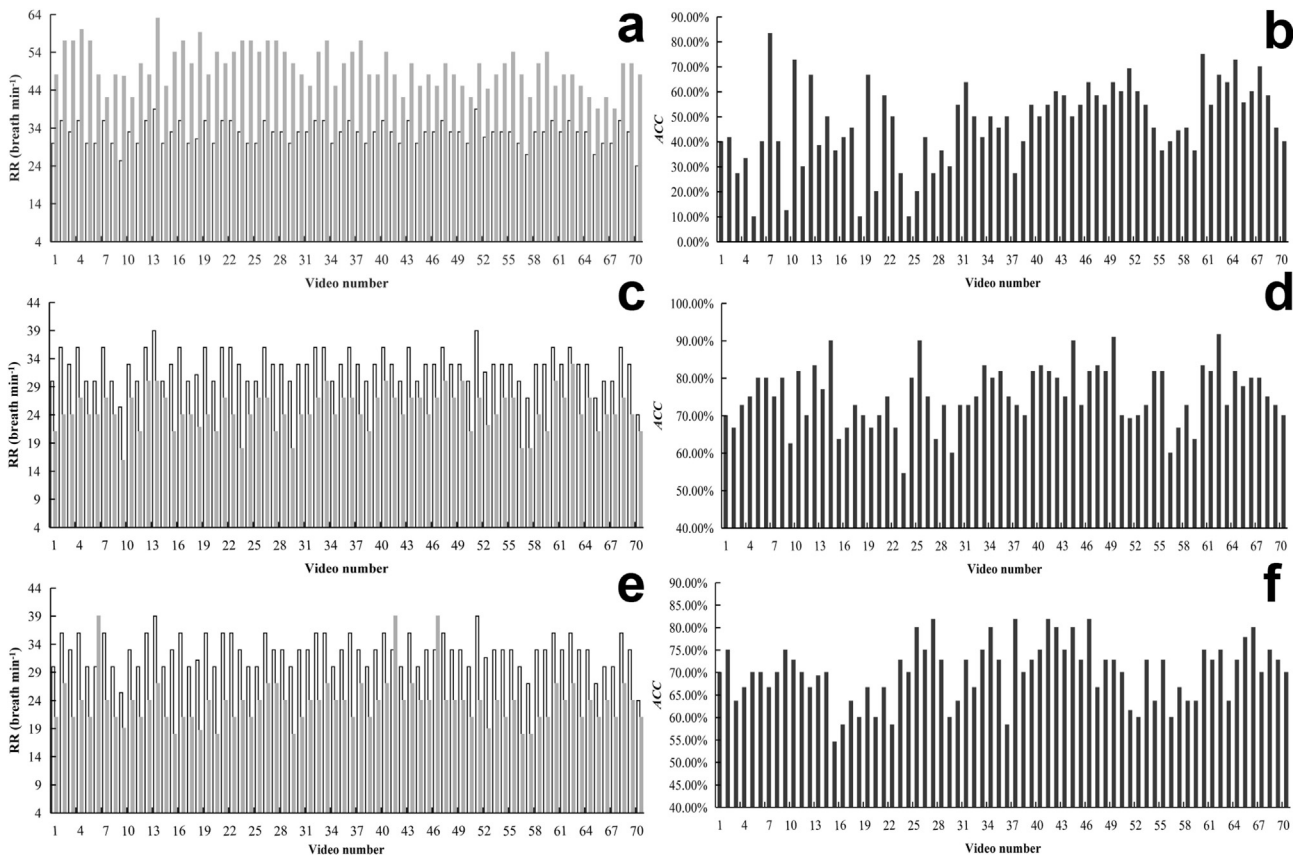


Fig. 13 – Detection results dairy cow RR with 3 methods under 70 test videos: including Method 1 (Figs a and b), Method 2 (Figs c and d) and Method 3 (Figs e and f), in Figs. a, c and e, the white histogram and the gray histogram respectively represent the actual RR of each video and the RR detected by the proposed method; Figs b, d and e respectively represent the RR detection accuracy of the three methods.

methods, and the average RR detection error was the lowest, indicating that it had better robustness.

In Method 1, it was difficult to detect the weak respiratory movement of a standing cow, and the respiratory movement optical flow caused by respiration was easily submerged by interference optical flow generated by other moving objects or self-movement, resulting in abnormal average optical flow direction and large detection error. Although Method 2 avoided the interference optical flow caused by other moving targets, since the accuracy of the target segmentation was not 100%, the segmentation would also generate some interference optical flow, which would influence the posterior probability detection results. Method 3 not only amplified respiratory movement, but also other movement interferences, so RR detection result was not ideal.

The errors of the three comparison methods and the proposed method mainly come from two aspects. One was the performance of different methods in detecting the respiratory movement of the standing resting dairy cow, which was the key to determine the accuracy of the RR detection of this method. The other was the error of the RR calculation method in this research. When calculating the RR, the respiratory cycle number of the cow was obtained by using Eq. (9) mainly according to the breathing direction curves detected by different methods, and finally the RR detected by different methods was calculated by combining the video duration and Eq. (11). Although it may be easy to miss an incomplete respiratory cycle at the end of the video, the error would not accumulate with time, and should not exceed one respiratory cycle. Therefore, this error will not greatly affect the RR detection results of different methods.

In summary, the proposed method in this study was the most suitable for the detection of the RR of standing resting cows.

4. Discussion

Since the dairy farming environment investigated in this study was complex and variable, and there were numerous interference factors that can affect the accurate detection of the breathing rate of standing resting cows, a number of these factors were analysed to verify the effectiveness of the proposed method. The method has been developed based on videos of standing resting cows, but the purpose is to be able

to detect changes in cows physiological state: the method was therefore applied to an available set of lying cows that were exposed to variable environments including known heat stress conditions.

4.1. Analysis of the effects of interference from strenuous movement on the detection of the RR of standing resting cows

Figure 14 shows an example of the failure to accurately detect RR in the presence of severe motion disturbances. It can be seen from Fig. 14a that the proposed method successfully detected two respiratory cycles before a vigorous movement by the cow target and six respiratory cycles after the movement, but failed to detect the two breaths during the activity, and the accuracy of RR detection in this video was only 80.00%.

As shown in Fig. 14a, two breaths are missed between points a and b. The main reason for this was that the optical flow disturbance caused by the intense physical movement after point a hid the change in the optical flow of the body's movement associated with normal breathing, resulting in weakening of the regularity of change of the average optical flow direction; the change in breathing direction was not obvious, leading to misdetection of RR and a decrease in detection accuracy.

Of the 70 test videos, 61 contained motion interference, and the average ACC of RR detection was 93.04%, which was higher than the 80.00% for Video 34. This was because the motion interference was not intense and global, the interfering optical flow was small, and the change in the average optical flow direction was not completely 'submerged' and did not influence the RR detection based on the change in the direction of the average optical flow.

In summary, even under conditions of strenuous motion, as long as the interference optical flow generated by the motion was smaller than the optical flow generated by the dairy cow's breathing movement, the impact on the accuracy of the RR detection was small, thereby proving that the proposed method has good robustness.

4.2. Impact of interference from other cows on the results of RR detection

The respiratory direction detection for Video 68, which contains interference from another cow, is shown in Fig. 15. In

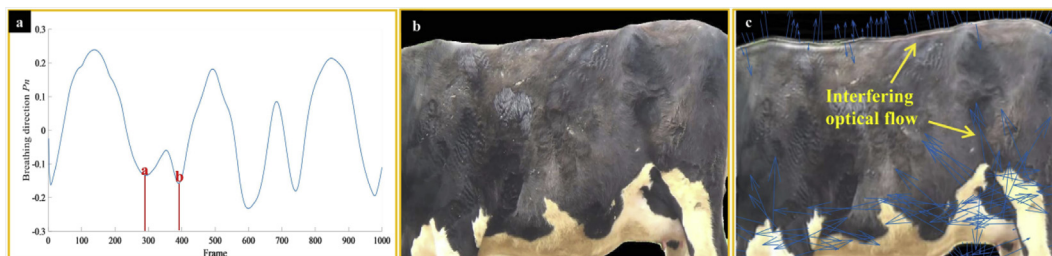


Fig. 14 – Proposed method erroneously detects the RR of cow undergoing strenuous movement in Video 34: (a) shows the breathing direction detected by the proposed method; (b) and (c) are respectively the cow target segmentation result and the optical flow detection result after breathing movement amplification; points a (the 290th frame) and b (the 390th frame) are the start and end positions of vigorous movement respectively.

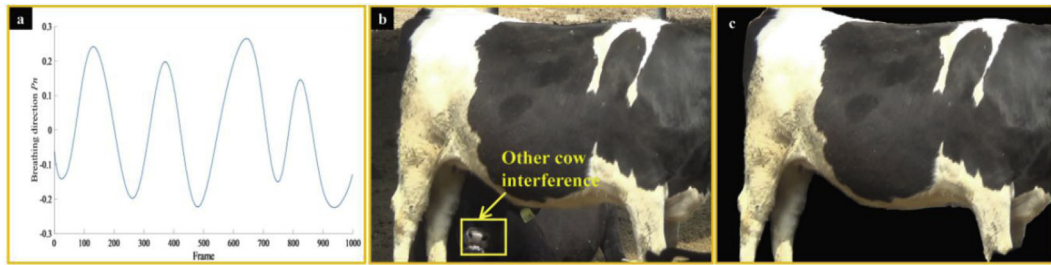


Fig. 15 – RR detection results of using proposed method in the presence of interference from other cows in Video 68: (a) breathing direction corresponding to Video 68; (b) and (c) are the original video with interference from other cow and the segmentation result of cow target respectively.

Figs. 15a and 8 respiratory cycles were successfully detected, and the actual number of respiratory cycles in Video 68 was 8, indicating that the detection accuracy of RR for this video was 100.00%. This was mainly because our method effectively removed irrelevant targets; moreover, since no other motion interference was present in the video, no extra interference was introduced into the optical flow in the results, and hence the accuracy of RR detection was improved.

Of the 70 test videos in this study, three videos contained interference from other cows. In these three videos, although the target cow's body was partially occluded by other cows, the interference from irrelevant targets could be removed after target segmentation, which effectively avoided errors in RR detection. This showed that the method could accurately detect the RR of standing resting cows, regardless of the interference from other cows.

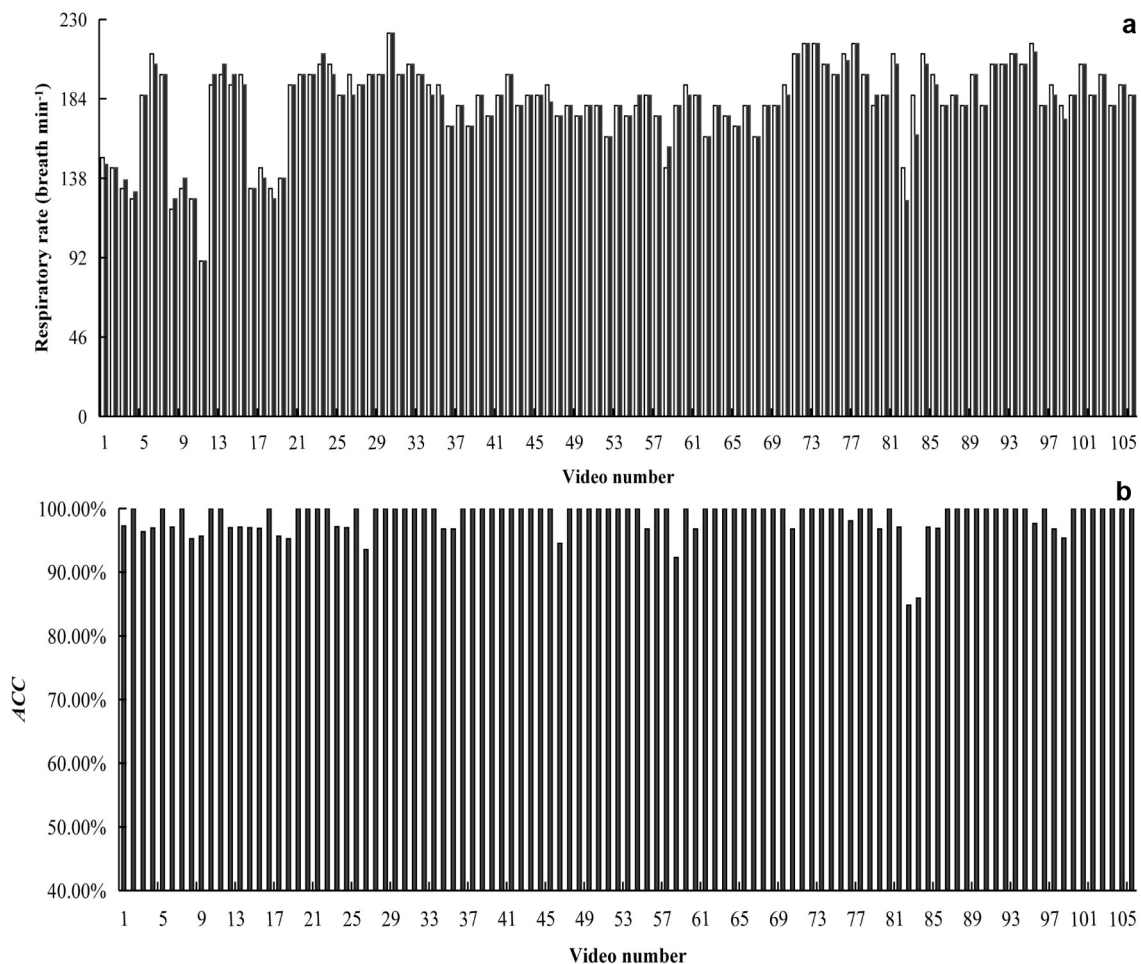


Fig. 16 – Detection results of cow RR under heat stress:(a) is the test results of each video, wherein the white histogram represents the actual RR of each video, gray histogram represents the RR detected by the proposed method; (b) is the accuracy of RR detection for each video.

4.3. Effects of heat stress on the results of RR detection

As the RR of dairy cows will increase sharply and the respiratory amplitude would decrease during heat stress, the detection results of the proposed method may be affected when applied to stressed cows. In order to verify the effectiveness of the method in detecting the RR of dairy cows under heat stress, 105 pieces of test videos under heat stress (from Song et al., 2019) were used, and the RR detection results are shown in Fig. 16. The actual RR was between 90 and 222 breath min^{-1} , and the detection result of the proposed method was between 90 and 222 breath min^{-1} . The average accuracy of the method for detecting the RR of cows under heat stress was 98.69%, which was 5.65% and 0.11% higher than that of the normal state in this study and the heat stress state in Song's study, respectively. The main reason was that the respiratory amplitude of lying cows was relatively large and the proposed method could amplify the weak respiratory movement, which could improve the accuracy for detecting weak respiration in the presence of interference to a certain extent. The results showed that the method could effectively detect the RR of dairy cows under heat stress even if the RR increased and the amplitude decreased.

4.4. Influence of breathing rate perturbation on the detection results

Since the upper limit of RR and the sampling frequency of 2-times accelerated processing meet Shannon sampling

theorem during severe heat stress of dairy cows, the original video information can be represented by 2-times accelerated video (Zhang, Wang, & Wang, 2017). In order to verify the detection effect of this method in the presence of breathing rate fluctuations, we accelerated the first 500 frames of each test video twice and left the rest unchanged to simulate breathing rate fluctuations, and then performed RR detection. The RR detection results are shown in Fig. 17. The actual average RR of 70 videos was 43.59 breath min^{-1} , and the average RR detected using the proposed method was 41.63 breath min^{-1} , and the average detection accuracy was 92.69%, which was only 0.35% lower than that of the original videos, indicating that the proposed method could still adequately detect the RR of standing resting cows well even when the RR fluctuated.

4.5. RR detection effects of the proposed method on lying cows

In order to verify the RR detection effect of the proposed method on lying cows, we used this method to detect the RR from 20 videos of lying cows. As shown in Fig. 18, the detection result of one of the videos showed that the actual RR of the cow in the video was 72 breath min^{-1} , the detection result was 72 breath min^{-1} , and the RR detection accuracy was 100%. In Fig. 18b, although this method could segment the lying cow target well, it mistakenly segmented part of the background into the cow target at the edge. This false segmentation may not have a significant impact on the RR detection results

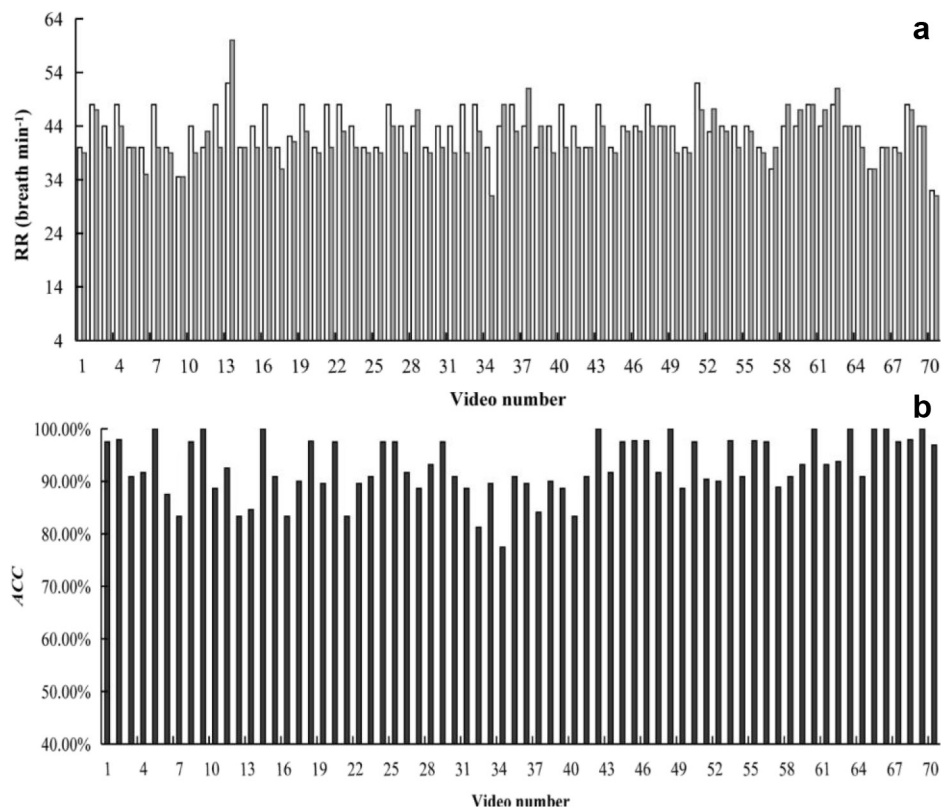


Fig. 17 – Detection results of the proposed method when respiratory rate has been artificially perturbed: (a) RR detection results of each video, where white histogram and gray histogram respectively represent the actual RR and the RR detected by the proposed method; (b) is the accuracy of RR detection for each video.

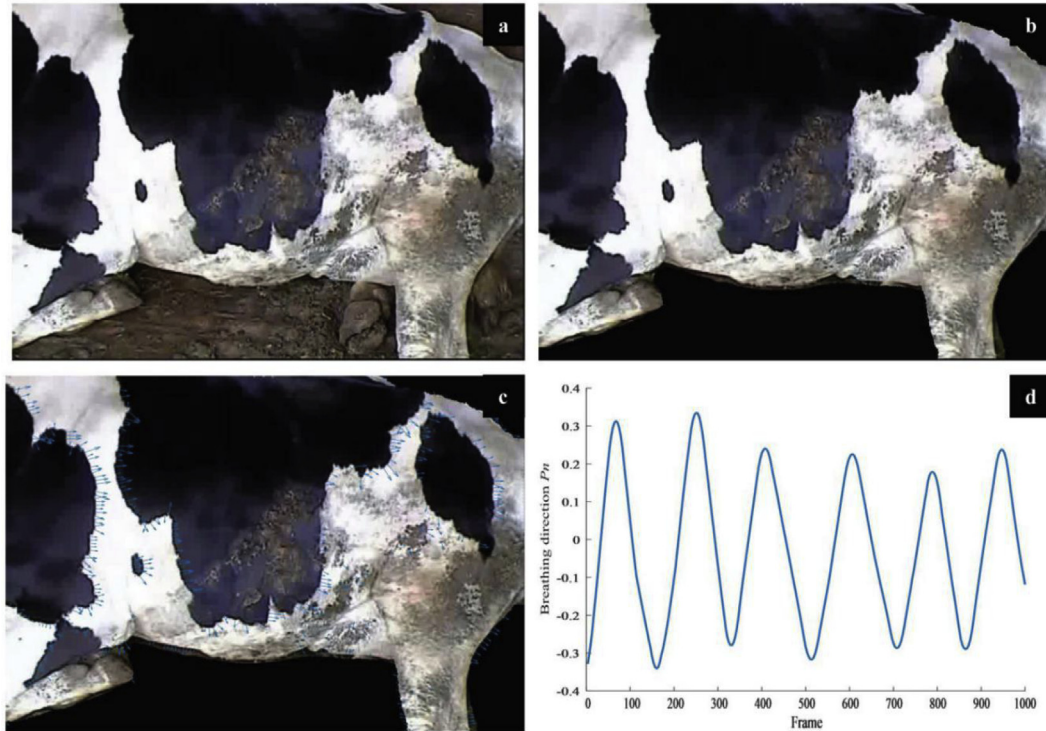


Fig. 18 – RR detection result of the lying cow by the proposed method: (a) is the original video of the lying cow, (b) is the corresponding cow target segmentation result, (c) is the optical flow detection result of the method, and (d) is the respiration direction detection result of the video.

because the respiratory motion amplitude of the lying cow was relatively large, and the PBVM algorithm adopted in this method only amplifies the motion within the respiratory frequency range, which can cover up to some extent interference optical flow caused by irregular changes of edge pixels. The average detection accuracy of RR in these 20 videos was 95.10%, which was 2.06% higher than that of standing dairy cows in this study. The results showed that this method can provide a certain reference for the detection of RR of lateral lying cows.

4.6. Effects of weather and wind on the results of RR detection

In this study, 70 test videos were used to test the proposed algorithm. The test videos include varying environmental conditions, including sunny, and cloudy, and different wind velocities. The test results showed that the average accuracy of the proposed algorithm was 92.86% and 93.01% for sunny and cloudy days, respectively, and its variance was 1.13×10^{-6} . The average accuracy for different wind velocities was 93.83%, 92.62% and 92.64%, respectively, and the variance was 0.5×10^{-4} . These results showed that weather changes and wind factors have little influence on the accuracy of the detection of RR for a standing dairy cows. The proposed algorithm could detect the RR with high accuracy, regardless of the environmental conditions.

5. Conclusion

For the intelligent detection of the respiratory rate (RR) of standing resting cows, we proposed a method involving deep learning and video analysis; 70 videos of standing resting dairy cows were obtained and tested under varying weather conditions and complex interference factors. The main conclusions are as follows:

- 1) The proposed method amplified the respiratory movement of the standing resting dairy cow on the basis of target segmentation, and the RR detection model was then used to calculate the RR of the standing cows by calculating the changes in the average optical flow direction in each frame. Except for severe disturbances, other interference did not have a significant impact on the final results. This also indicated that using the proposed method to detect the breathing rate of standing cows was possible.
- 2) The PBVM algorithm was introduced for cow target segmentation, which not only amplified the respiratory movement of standing cows, but also reduced the interference of irrelevant background movement on breath detection, thereby improving the reliability of the proposed method. The accuracy of respiratory detection of our method was 22.91% higher and the

average error was 7.2 breath min^{-1} , lower than those of Method 3, indicating the effectiveness of the PBVM algorithm in amplifying the respiratory movement of standing resting cows.

- 3) The method proposed in this study can be used for the accurate detection of the RR of standing resting cows. The test results showed that the average accuracy was 93.04%, and the average error was 2.4 breath min^{-1} . This research could provide a valuable basis for automatic monitoring of heat stress and remote diagnosis of respiratory-related diseases in dairy cows.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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